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format short eng
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```
%Problem 1
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```
f = 33e9;  
c = 2.998e8;  
er = 2.2;
```

```
% Wavelength
```

```
lambda = c / (f * sqrt(er))
```

```
lambda =  
6.1250e-003
```

```
% Minimum
```

```
min = lambda / 10
```

```
min =  
612.5004e-006
```

Problem 5

a) If the line is infinitely long it never reflects back meaning the result is 75 Ohms. Since its infinitely long the resistance measures the characteristic impedance.

b) Since its open and the distributed resistance is low it would result in infinite resistance. basically an open circuit.

```
%Problem 7
```

```
t = linspace(0,17e-6,1000);
```

```
Vplus = 50;  
Vminus = -2/3 * Vplus;
```

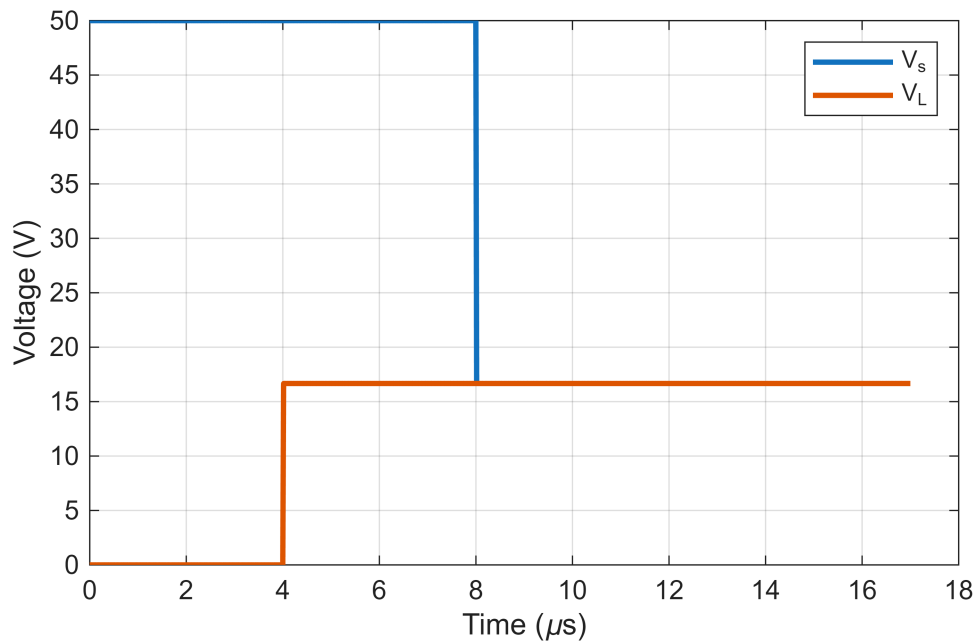
```
Vs = Vplus*(t >= 0) + Vminus*(t >= 8e-6);  
VL = (Vplus + Vminus)*(t >= 4e-6);
```

```
figure;  
plot(t*1e6, Vs, 'LineWidth',2); hold on;  
plot(t*1e6, VL, 'LineWidth',2);
```

```

xlabel('Time (\mus)');
ylabel('Voltage (V)');
legend('V_s','V_L');
grid on;

```



```

%Problem 8
t = linspace(0,17e-6,1000);
Vplus = 50;

figure;

% part a
Vminus = (1/3)*Vplus;

Vs = Vplus*(t >= 0) + Vminus*(t >= 8e-6);
VL = (Vplus + Vminus)*(t >= 4e-6);

subplot(3,1,1);
plot(t*1e6, Vs, 'LineWidth',2); hold on;
plot(t*1e6, VL, 'LineWidth',2);
title('(a) R_L = 100 \Omega');
ylabel('V');
legend('V_s','V_L');
grid on;

% part b
Vminus = -Vplus;

Vs = Vplus*(t >= 0) + Vminus*(t >= 8e-6);

```

```

VL = (Vplus + Vminus)*(t >= 4e-6);

subplot(3,1,2);
plot(t*1e6, Vs, 'LineWidth',2); hold on;
plot(t*1e6, VL, 'LineWidth',2);
title('(b) R_L = 0');
ylabel('V');
legend('V_s', 'V_L');
grid on;

% part c
Vminus = Vplus;

Vs = Vplus*(t >= 0) + Vminus*(t >= 8e-6);
VL = (Vplus + Vminus)*(t >= 4e-6);

subplot(3,1,3);
plot(t*1e6, Vs, 'LineWidth',2); hold on;
plot(t*1e6, VL, 'LineWidth',2);
title('(c) R_L = \infty');
xlabel('Time (\mu s)');
ylabel('V');
legend('V_s', 'V_L');
grid on;

```

